Nitobond AR

constructive solutions

Acrylic emulsion cement modifier and water based concrete bonding agent

Uses

Recommended primer for Fosroc cementitious repair system. For improving and bonding floor toppings, renders and mortars; repair of worn, damaged or spalled concrete; polymer modified floor screeds.

Advantages

Simply brush onto concrete as a primer. Single component, gauged as required.

- High adhesion -Provides excellent bond to concrete, masonry, stone work plaster and block board.
- Increased strength - Improves tensile and flexural properties of mortar. Allows thin section applications.
- Versatile - Compatible with all common hydraulic cements
- Economical - Economical to use

Description

Nitobond AR is a modified acrylic emulsion specially designed for use as a bonding aid and gauging liquid for cementitious systems. It is resistant to hydrolysis and can therefore be used for external applications.

Nitobond AR is the recommended primer for the Fosroc cementitious repair system.

Technical support

Fosroc offers a technical support service to specifiers, end users and contractors, as well as on-site technical assistance in locations all over the country.

Properties

Specific gravity : 1.025 to 1.04

Mechanical properties

Typical improvements in mechanical properties of a 3 : 1 cement mortar /sand using Nitobond AR.

Mechanical properties	Curing conditions	Control	Nitobond AR
Adhesion to concrete Slant shear strength N/mm^2 (BS 6319)	Dry	5.0	20.0

Note: The typical physical properties given above are derived from testing in a controlled laboratory environment. Resultsderived from testing field-applied samples may vary, dependent on actual site conditions.

GreenPro Ecolabel

Specification clauses

As a bonding agent

The bonding agent shall be Nitobond AR an acrylic based emulsified solution containing a minimum of 43% solids and compatible with cementitious materials. The bonding agent shall provide adequate bond strength when directly applied on concrete and also mixed with neat cement.

As a mortar modifier

The mortar modifier shall be Nitobond AR, an acrylic emulsified cement modifier which provides good bond with concrete / masonry and improves the flexural strength, tensile strength properties of the mortar when added in the specified quantities. It shall be compatible with all common hydraulic cements.

Application instructions

Surface Preparation

The object of the surface preparation is to achieve a clean sound surface with a good mechanical key. All substrates should be cleaned and free of dust, plaster, oil, paint, grease, corrosion deposits, and any other deleterious substances. Laitence should be removed by mechanical means. Oil or greasy deposits should be removed by suitable means. All surfaces so treated should be thoroughly washed with clean water. Smooth substrates must be mechanically roughened e.g. by scabbling, needle gun or grit/ sand blasting to provide a mechanical key.

Concrete repairs : Remove all surface dust and debris. Any reinforcing steel in the repair area should be exposed around its full circumference and cleaned to remove all loose scale and corrosion deposits and then primed with Nitozinc Primer.

Priming

Immediately before priming, the concrete substrate should be thoroughly dampened with water and any excess being brushed off. Scrub Nitobond AR into the substrate. Avoid pudding of the emulsion. The repair mortar/topping should be applied whilst the primer is still tacky. The bond coat becomes tacky in about 2 minutes and best results are achieved if the mortar is applied within the next 5-10 min.

In case of Nitobond AR slurry coat (1 volume of Ntiobond AR + 1.5 volume of water + 1.5 volume of cement), best results are achieved, if mortar is applied within 5 minutes of application of bond coat.

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Typical Mix designs to modify mortars :

OPC	50 kg
Zone 2 sand	150 kg
Nitobond AR	8 - 10 litres
Water	Add sufficient water to give required consistency

Note : Trials are recommended to optimise mix designs. 1:4 cement sand mortars can also be used with same dosage of Nitobond AR.

General recommendations

For repairs

Prepare surfaces thoroughly as previously recommended Cut back edges to avoid feather edging. Prime all surfaces including edges, using Nitobond AR.

All applications shall be wet on wet. For best results Nitobond AR should be allowed to become tacky. If the bonding agent is allowed to dry for more than 15 minutes at 30^0 C initial 'grab' to the repair mortar will be reduced. Therefore, the use of temporary shuttering is recommended. Water content of the mortar should be kept to the minimum necessary.

For consistent results, the use of clean, dry sand is recommended. Where the use of wet sand is unavoidable, the quantity of water to be added must be reduced.

Cure mortars using Nitobond AR to prevent rapid drying out.

Protect uncured mortar from frost and rain. Do not retemper mortar after initial set.

Minimum application temperature for Nitobond AR is 10^0C, but the mortar should not be applied if the temperature is expected to fall.

For permanently immersed conditions consult Local Fosroc office.

Limitations

As a bonding agent

Nitobond AR may exhibit less overlay time at higher temperature. In such cases as overlay mortar shall not be applied when Nitobond AR is totally dry.

Nitobond AR when used as bonding agent cannot act as a barrier coat against ingress of chloride ions from the substrate.

Equipment cleaning

Immediately after use, wash all tools with clean water.

Estimating

Packaging

Nitobond AR is supplied in 1, 5, 20, 25, 50, 100 and 200 litre plastic containers..

Coverage

Approximately 6 - 8m^2/ litre depending on substrate

Storage

Shelf life

12 months at 30^0C. Nitobond AR should be protected from frost.

Precautions

Health & Safety instructions

Nitobond AR is non toxic. However it should never be ingested and if it comes in contact with eyes, wash immediately with plenty of water and seek medical treatment.

Nitobond AR is slightly alkaline. Skin contact should be avoided. Gloves and protective clothing should be worn during application.

Fire

Nitobond AR is non flammable.

Additional information

Nitobond AR forms a part of the Fosroc concrete repair system.

Fosroc manufactures a wide range of products specifically designed for repair and refurbishment of damaged reinforced concrete. This includes hand-placed and spray grade repair mortars, fluid micro concretes, chemical resistant epoxy mortars and a comprehensive package of protective coatings. In addition, a wide range of complementary products is available. This includes joint sealants, water proofing membranes, grouting, anchoring and specialised flooring materials.

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Important note :
Fosroc products are guaranteed against defective materials and manufacture and are sold subject to its standard terms and conditions of sale, copies of which may be obtained on request. Whilst Fosroc endeavours to ensure that any advice, recommendation specification or information it may give is accurate and correct, it cannot, because it has, no direct or continuous control over where or how its products are applied, accept any liability either directly or indirectly arising from the use of its products whether or not in accordance with any advice, specification, recommendation or information given by it.

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